

2026-08-21 – System Design – Metrics Monitoring



What is a Metrics Monitoring Platform?

A metrics monitoring platform collects performance data (CPU, memory, throughput, latency) from servers and services, stores it as time-series data, visualizes it on dashboards, and triggers alerts when thresholds are breached. Think Datadog, Prometheus/Grafana, or AWS CloudWatch. This is infrastructure that engineers rely on to understand system health and respond to incidents.

Functional Requirements

Core Requirements

1. The platform should be able to ingest metrics (CPU, memory, latency, custom counters) from services
2. Users should be able to query and visualize metrics on dashboards with filters, aggregations, and time ranges
3. Users should be able to define alert rules with thresholds over time windows (e.g., "alert if p99 latency > 500ms for 5 minutes")
4. Users should receive notifications when alerts fire (email, Slack, PagerDuty)

Below the line (out of scope):

- Log aggregation and full-text search (separate concern)
- Distributed tracing (spans, traces)
- Anomaly detection via ML

Non-Functional Requirements

Core Requirements

1. The system should scale to ingest 5M metrics per second from 500k servers
2. Dashboard queries should return within seconds, even for queries spanning days or weeks
3. Alerts should evaluate with low latency (< 1 minute from metric emission to alert firing)
4. The system should be highly available. We can tolerate eventual consistency for dashboards, but alert evaluation should be reliable.
5. The system should handle late or out-of-order data gracefully (network delays are common)

Below the line (out of scope):

- Multi-region replication (would add complexity)
- Strong consistency guarantees